

Mixed Practice

A. Second Largest Element

1 second, 256 megabytes

You are given an integer array A of size n .

Your task is to find the second largest **distinct** element present in the array.

If the second largest distinct element does not exist, print -1 .

Input

The first line contains an integer n ($0 \leq n \leq 100$).

The second line contains n integers A_i ($1 \leq A_i \leq 1000$).

Output

Print the second largest distinct element in the array. If it does not exist, print -1 .

input
5 4 3 10 9 2
output
9

input
7 13 6 31 14 29 44 3
output
31

input
4 5 5 5 5
output
-1

In the first example:

- The array is: $[4, 3, 10, 9, 2]$.
- The largest element is 10 .
- The next largest **distinct** element is 9 .

Therefore, the second largest element is 9 .

Note that we consider distinct values. For example, if the array were $[5, 5, 5]$, there would be no second largest element, and the answer would be -1 .

B. Push Zeros to the End

1 second, 256 megabytes

You are given an integer array for multiple test cases.

For each test case, move all zero elements to the end of the array while maintaining the relative order of the non-zero elements.

Input

The first line contains an integer t ($1 \leq t \leq 100$) — the number of test cases.

For each test case:

- The first line contains an integer n ($0 \leq n \leq 10^5$).

- The second line contains n integers A_i ($-10^9 \leq A_i \leq 10^9$).

The sum of n over all test cases does not exceed 10^5 .

Output

For each test case, print the modified array after pushing all zeros to the end.

input
1 7 2 0 0 1 3 0 0
output
2 1 3 0 0 0 0

input
2 5 0 3 0 2 0 5 9 0 0 8 2
output
3 2 0 0 0 9 8 2 0 0

In the first example, the array is:

[2 0 0 1 3 0 0]

After moving all zeros to the end while preserving the order of non-zero elements:

- Non-zero elements remain in the same order: 2, 1, 3
- All zeros are moved to the end.

Result:

[2 1 3 0 0 0 0]

C. Check array rotation

1 second, 256 megabytes

You are given a sorted array that has been rotated to the right by some number K ($K \geq 0$).

In other words, the array was originally sorted in increasing order, and then the last K elements were moved to the front.

Your task is to determine the value of K , i.e., the index of the smallest element in the array.

Input

The first line contains an integer t ($1 \leq t \leq 100$) — the number of test cases.

For each test case:

- The first line contains an integer n ($2 \leq n \leq 10^5$).
- The second line contains n integers A_i ($-10^9 \leq A_i \leq 10^9$).

Each array is obtained by rotating a sorted array.

The sum of n over all test cases does not exceed 10^5 .

Output

For each test case, print a single integer — the value of K .

input
1 6 5 6 1 2 3 4
output
2

input
2 5 3 6 8 9 10 4 10 20 30 1
output
0 3

In the first example:

- The array is: [5 6 1 2 3 4].
- The original sorted array would be: [1 2 3 4 5 6].
- After rotating the array to the right, the last elements move to the front.
- The smallest element (1) appears at index 2 (0-based indexing).

Therefore, the array has been rotated by $K = 2$.

If the smallest element appears at index 0, it means the array was not rotated.

D. Rotate Array

2 s, 256 megabytes

You are given an integer array for multiple test cases.

For each test case, rotate the array to the left by D positions.

A left rotation by D moves the first D elements of the array to the end while preserving their order.

Input

The first line contains an integer t ($1 \leq t \leq 10^4$) — the number of test cases.

For each test case:

- The first line contains an integer n ($0 \leq n \leq 10^6$).
- The second line contains n integers A_i ($-10^9 \leq A_i \leq 10^9$).
- The third line contains an integer D ($0 \leq D \leq n$).

The sum of n over all test cases does not exceed 10^6 .

Output

For each test case, print the rotated array.

input
1 7 1 2 3 4 5 6 7 2
output
3 4 5 6 7 1 2

input
2 7 1 2 3 4 5 6 7 0 4 1 2 3 4 2
output
1 2 3 4 5 6 7 3 4 1 2

In the first example, the array is [1 2 3 4 5 6 7].

After rotating left by $D = 2$:

- the first two elements (1, 2) move to the end,
- the remaining elements shift to the front.

The resulting array is:

A character is compressed only when its consecutive count is greater than 1.

Input

The first line contains an integer T ($1 \leq T \leq 5$) – the number of test cases.

Each of the next T lines contains a string S ($1 \leq |S| \leq 5000$).

It is guaranteed that the consecutive count of any character in the string does not exceed 9.

Output

For each test case, print the compressed string.

input
2 aaabddccc ggttttffffreee
output
a3b2d2c3 g2t4f4r2e3

```
2
aaabbbddccc
ggttttffffrreee
```

input
10000000000000000000000
output
4

output

a3b2d2c3
g2t4f4r2e3

G. Longest Word

1 second, 256 megabytes

You are given a string S consisting of English letters and spaces.

A *word* is a maximal contiguous sequence of letters. It is guaranteed that words are separated by exactly one space, and the first character of S is a letter.

Your task is to find the length of the longest word in S .

- ## Input

The only line contains the string S .

It is guaranteed that $1 \leq |S| \leq 10^5$, and S consists only of English letters and spaces.

Output

Print one integer — the length of the longest word in S .

input
hello world this is CP
output
5

H. Diagonal Sum

1 second, 256 megabytes

You are given a square integer matrix of size $n \times n$.

Your task is to compute the sum of the main diagonal (top-left to bottom-right).

Input

The first line contains a single integer n ($1 \leq n \leq 1000$) — the size of the matrix.

The next n lines each contain n integers $a_{i,j}$ ($0 \leq a_{i,j} \leq 10^5$), representing the matrix.

Output

Print a single integer, the sum of the main diagonal elements.

input
3 1 2 3 4 5 6 7 8 9
output
15

$1 + 5 + 9 = 15$

I. Rotate Image

1 second, 256 megabytes

You are given a square matrix of size $n \times n$ filled with integers. Your task is to rotate the matrix by 90° in the **clockwise** direction.

Input

The first line contains a single integer n ($1 \leq n \leq 1000$), the size of the matrix.

Each of the next n lines contains n integers $a_{i,j}$ ($1 \leq a_{i,j} \leq 10^9$), the elements of the matrix.

Output

Print the rotated matrix.

input
3 1 2 3 4 5 6 7 8 9
output
7 4 1 8 5 2 9 6 3

input
5 12 47 3 88 65 29 54 76 11 92 7 81 34 56 23 90 14 67 45 38 19 72 8 99 50

output
19 90 7 29 12 72 14 81 54 47 8 67 34 76 3 99 45 56 11 88 50 38 23 92 65

input
3 4 1 2 3 4 5 6 7 8 9 10 11 12
output
1 2 3 4 8 12 11 10 9 5 6 7

J. Spiral Matrix

1 second, 256 megabytes

You are given an integer matrix with m rows and n columns.

Print all elements of the matrix in **clockwise spiral order** starting from the top-left cell: go right, then down, then left, then up, and repeat while shrinking the remaining rectangle.

Input

The first line contains two integers m and n ($1 \leq m, n \leq 10$).

Each of the next m lines contains n integers, the matrix elements ($-100 \leq a_{i,j} \leq 100$).

Output

Print $m \cdot n$ integers in one line: the elements in spiral order, separated by single spaces.

input
3 3 1 2 3 4 5 6 7 8 9
output
1 2 3 6 9 8 7 4 5